

Program 1

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ZZJOB 5                      CSC 001    GERALD, C. F.
ZZFORXS
*LIST PRINTER
C PROGRAM FOR SIMPSONS RULE EVALUATION OF SINE INTEGRAL BETWEEN X = 0 AND X = B
  F(X) = SIN(X)/X
C READ UPPER LIMIT OF INTEGRATION AND NUMBER OF PANELS, N GREATER THAN 2
  READ 100, B, N
C CHECK THAT NUMBER OF PANELS IS EVEN
  NHALF = N/2
  IF (N - 2*NHALF) 99,1,99
99 PRINT 200
  CALL EXIT
C COMPUTE DELTA X. ADD FIRST, SECOND, AND LAST VALUES TO SUM
  1 XN = N
  DELX = B/XN
  SUM = 1.0 + 4.*F(DELX) + F(B)
C ADD REST OF VALUES TO SUM
  X = 2.*DELX
  DO 10 I = 2, NHALF
    SUM = SUM + 2.*F(X) + 4.*F(X+DELX)
  10 X = X + 2.*DELX
C COMPUTE INTEGRAL AND OUTPUT THE VALUE
  SININT = DELX/3.*SUM
  PRINT 201, B, SININT, N
  CALL EXIT
100 FORMAT (F10.0, I10)
200 FORMAT (60H WITH ODD NUMBER OF PANELS, SIMPSONS 1/3 RULE DOES NOT
1 APPLY )
201 FORMAT (41H VALUE OF SINE INTEGRAL FROM X = 0 TO X = , F6.2,
1 3H IS , F10.7, 5H WITH , I4, 7H PANELS )
  END
5.0      100
ZZZZ

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VALUE OF SINE INTEGRAL FROM X = 0 TO X = 5.00 IS 1.5499307 WITH 100 PANELS

Note that if N should be odd, $NHALF$ will be less than exactly one-half of N because the remainder is truncated in the fixed point division.

After the first, second, and last terms in Simpson's rule sum are accumulated, the pattern of the remaining terms is repetitive so that a DO loop can readily accumulate the rest of the summation. After multiplying by $\Delta x/3$, the integral according to Eq. (4.2) results.

The second integration program (Program 2) uses Romberg integration with 10 subintervals arbitrarily chosen as the starting point. Again the sine integral is computed in the example. After a first approximation has been found with 10 subintervals, the value of Δx is halved, an improved approximation is found, and an extrapolated value is determined utilizing Eq. (2.2) with $n = 2$ the first time. If the difference between the integral with 20 panels and the extrapolated value exceeds an error tolerance, TOL, the interval halving, extrapolation, and comparing continues until either the error tolerance is met, or else eight halvings have been made, whereupon the process is abandoned. The output of the program is a table of successive approximations and corresponding successive extrapolations, together with a statement of the final result.